DENDROCHRONOLOGY

Tree growth, wood anatomy and tree-ring structure

Lesson 2 · 90 minutes · Reading rings through observation

Learning objectives

Locate wood production

Explain the vascular cambium and secondary growth in a diagram.

Read boundaries from anatomy

Compare softwood, ring-porous and diffuse-porous wood using visible features.

Interpret a ring count

Consider sampling height, missing pith, false and discontinuous rings.

Review: what event does a ring date?

Year of ring formation

Link this to felling or construction using additional evidence.

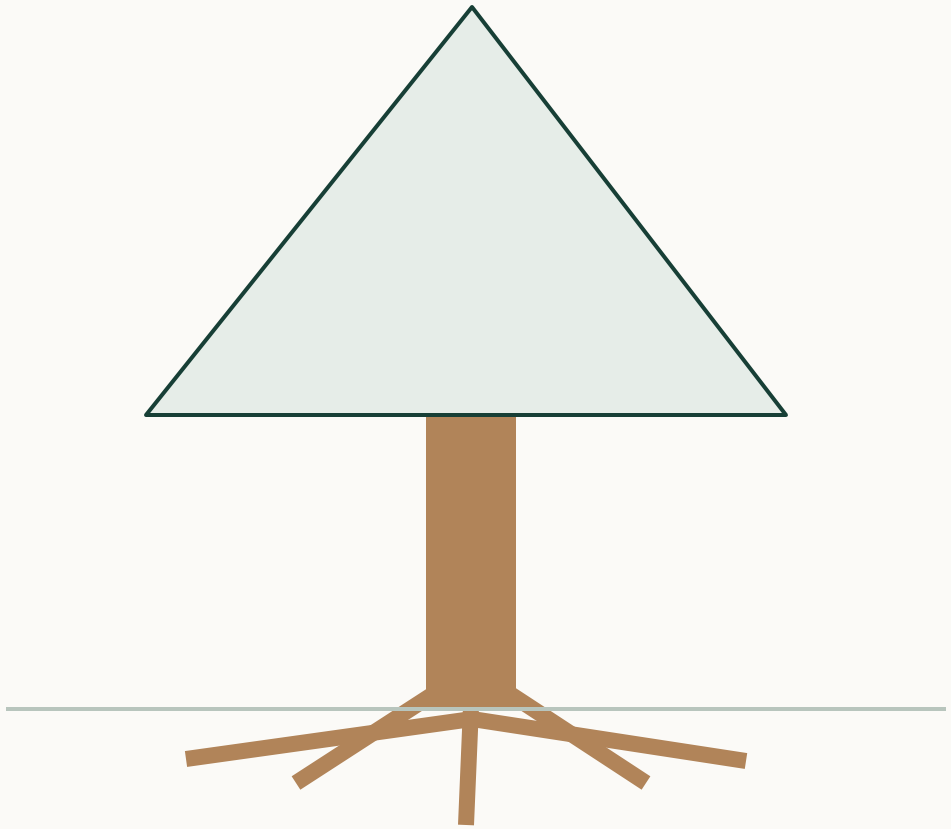
Annual increments

Check whether a growth ring represents one year.

Question for today

Will simply counting dark lines give a correct date?

A tree is a connected living organism



Leaves: photosynthesis and gas exchange

Stem: support, transport and storage

Roots: water and nutrient uptake

Original schematic; not a specimen or measurement, and not to scale.

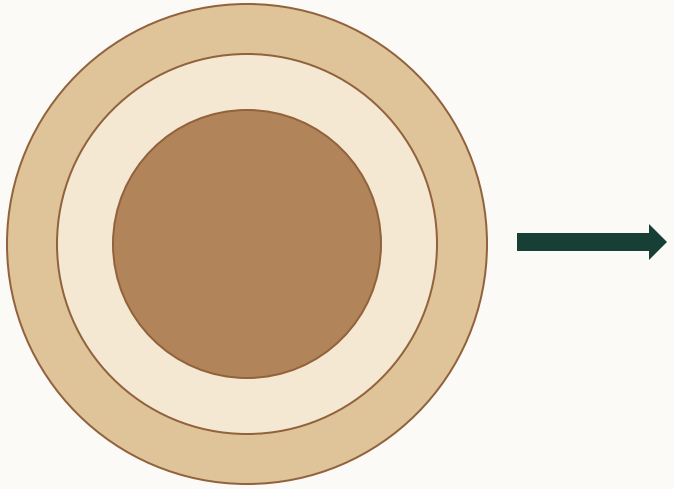
Growth in length and thickness

Primary growth
Apical meristems extend shoots and roots.



Extension at the tip

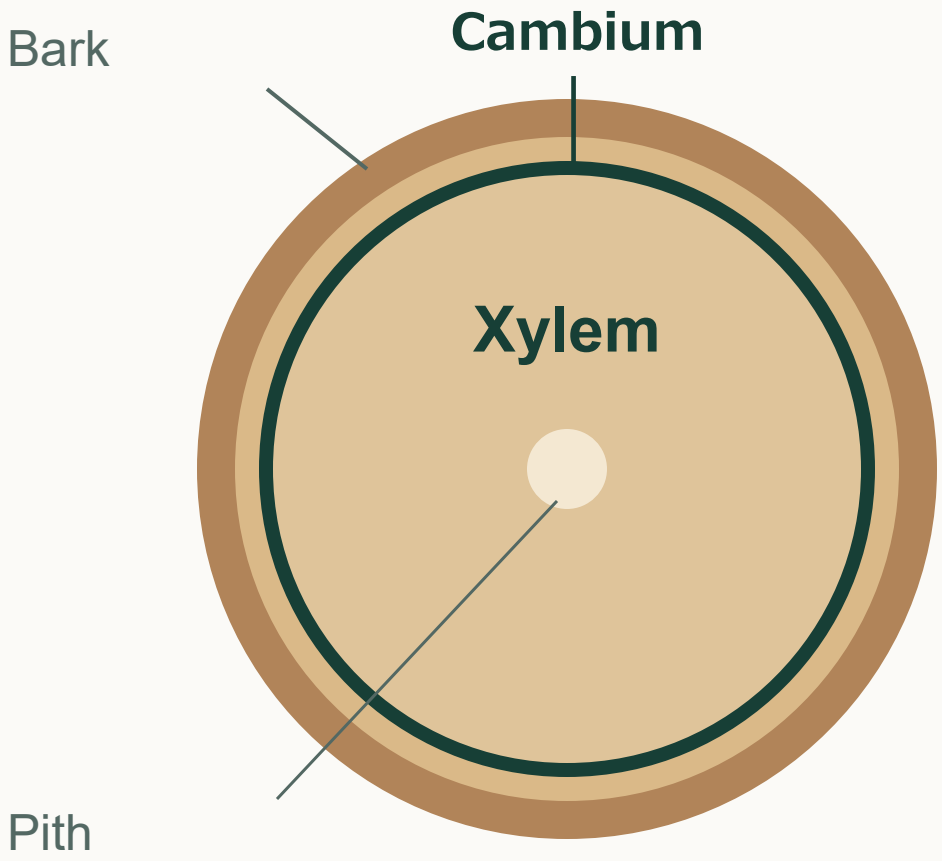
Secondary growth
Cambial activity adds wood and thickness.



New wood accumulates outward

Original schematic; not a specimen or measurement, and not to scale.

The cambium produces wood



Inward: secondary xylem

Outward: secondary phloem

The cambium lies between xylem and phloem.

Original schematic. Cambium (green) is exaggerated in thickness.

Wood also contains living cells

Tissue or cell	Typical state and function at maturity
Tracheids / vessel elements	Usually dead; involved in water conduction and support.
Sapwood parenchyma	Includes living cells for storage and metabolism.
Cambial cells	Living meristem supplying xylem and phloem cells.

Sapwood and heartwood



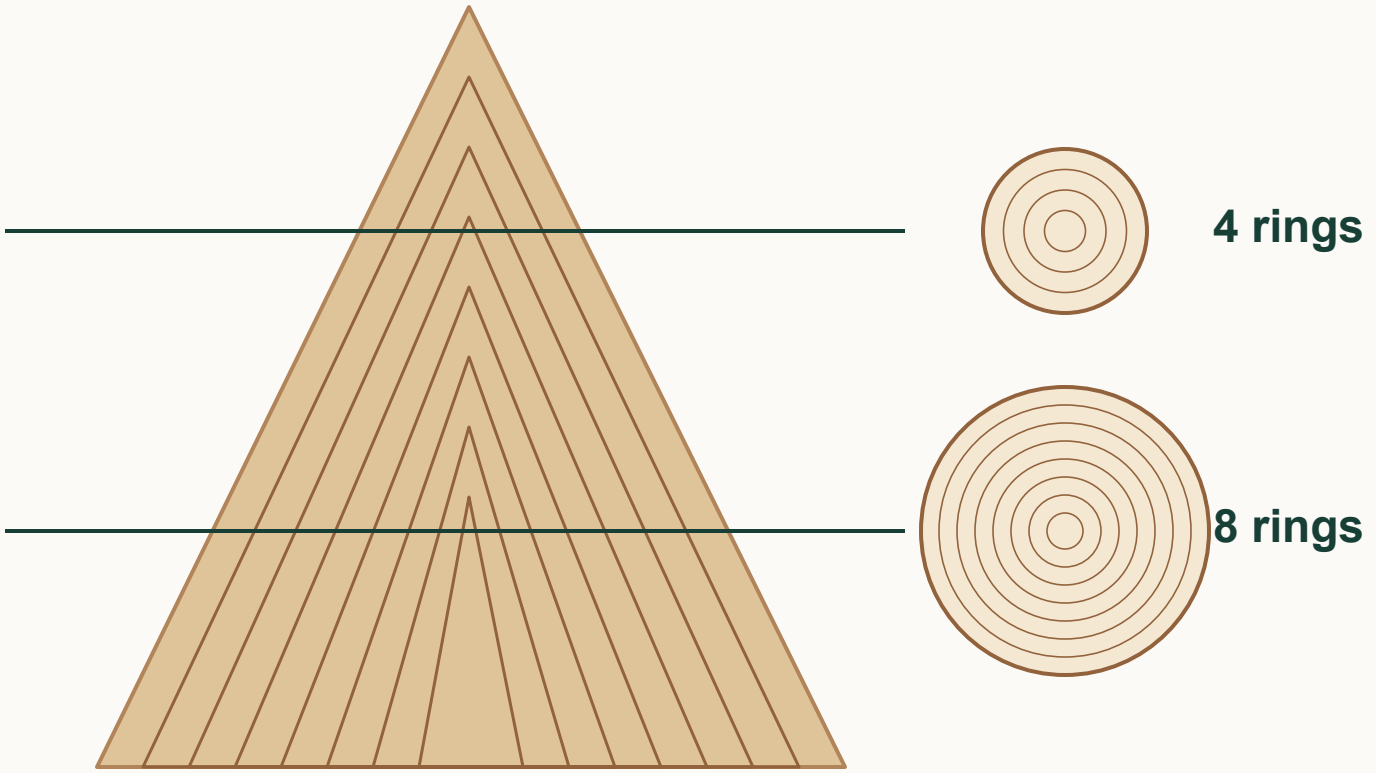
Sapwood
Outer wood involved in water transport and storage.

Heartwood
Inner, transformed wood, usually no longer conducting water.

A colour change is not itself a growth-ring boundary.

Original schematic; not a specimen or measurement, and not to scale.

Ring counts depend on sampling height



Upper stem sections form later.

A higher sample can contain a shorter record.

Distinguish a ring count from age since germination.

Idealised teaching example. Disk sizes and ring counts are not measurements.

Seasonality and wood formation

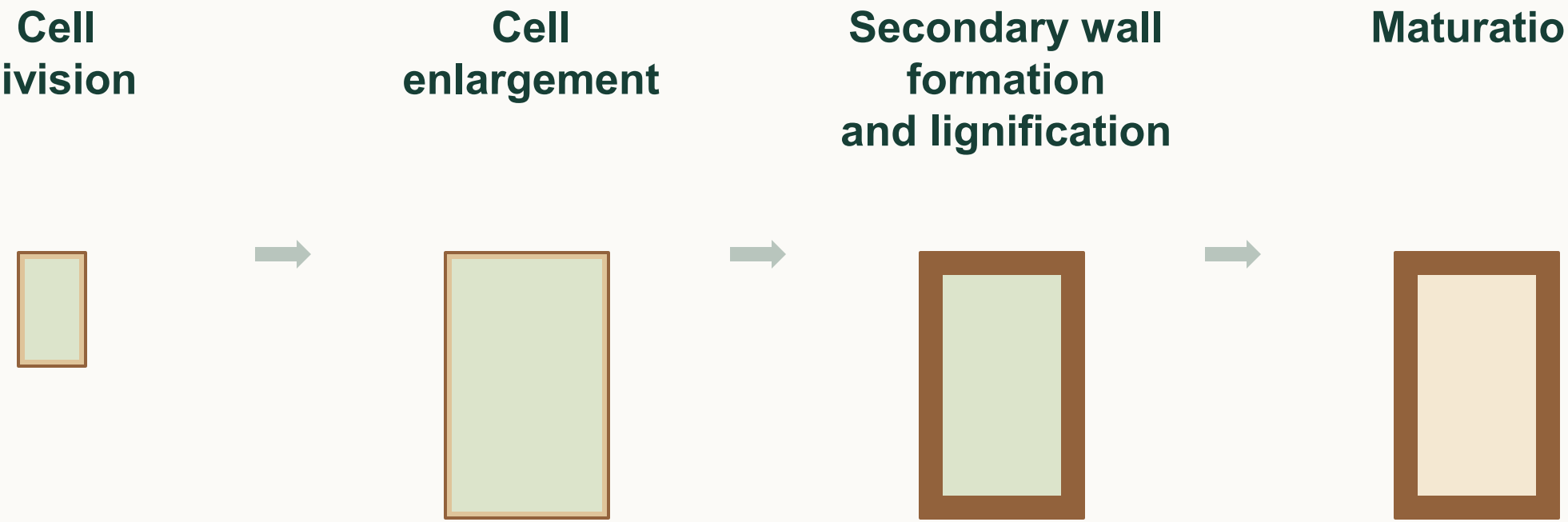
Dormancy and reactivation → **Cell production and maturation** → **Activity declines toward dormancy**



The timing and duration of wood formation vary with species and environment.

Schematic seasonal sequence. The horizontal axis does not show specific months or durations.

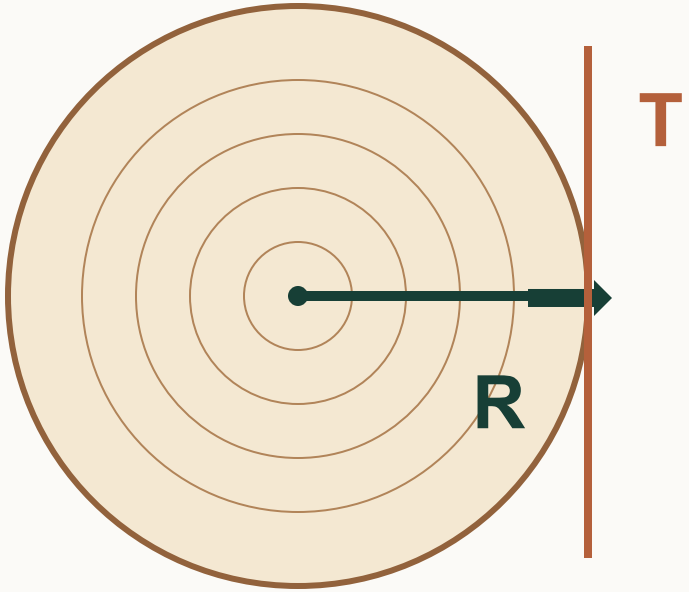
From cell production to mature wood



Different cells pass through overlapping developmental stages.

Original schematic; not a specimen or measurement, and not to scale.

The three directions: L, R and T



Looking at the transverse plane

Original schematic; not a specimen or measurement, and not to scale.

L: longitudinal

R: radial

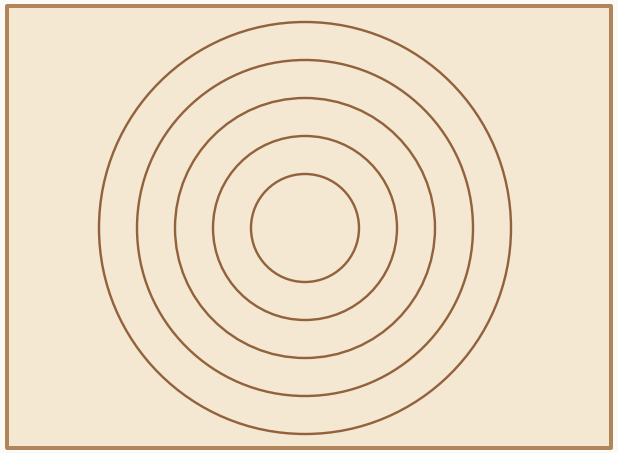
T: tangential

L is perpendicular to this drawing.

Rays extend in the R direction.

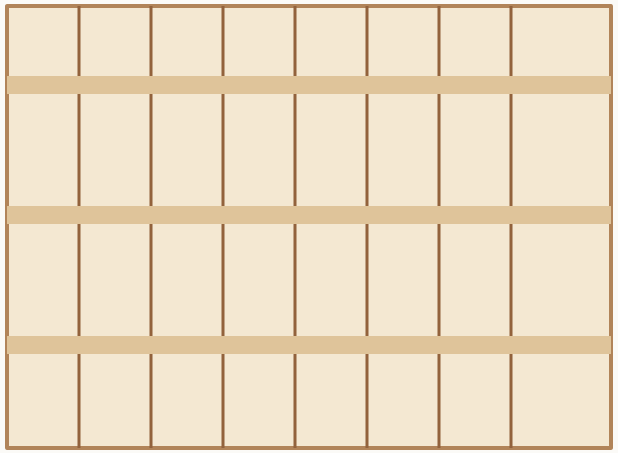
Three planes reveal the same tissue

Transverse section
R–T plane, perpendicular to L



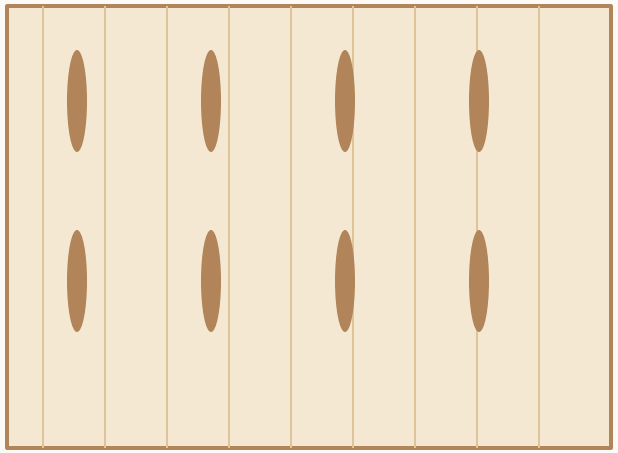
R → T ↑

Radial section
L–R plane



R → L ↑

Tangential section
L–T plane

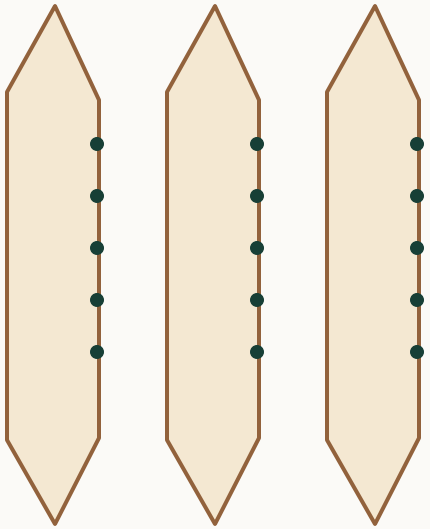


T → L ↑

Original schematic; not a specimen or measurement, and not to scale.

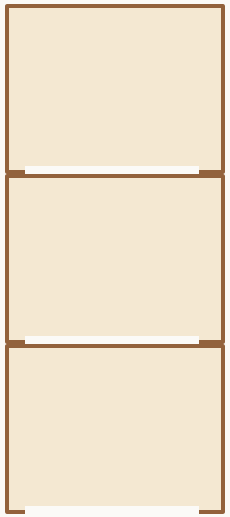
The shape of water-conducting cells

Tracheid
An elongated cell linked to neighbours through pits.



Pits

Vessel
A chain of vessel elements connected through perforations.



Connected through perforations

Original schematic; not a specimen or measurement, and not to scale.

Softwood: changes in tracheids

Observe

Look at cell rows and bands across rings.

Earlywood to latewood

Compare lumen size and wall thickness.

Ring boundary

Locate the transition to the next earlywood.

Open the specimen image online

Observe the image at the link

FFPRI: specimen record

Ring-porous wood: earlywood vessels

Observe

Find bands of large vessels.

Trace a ring

Compare the vessel band and surrounding tissue.

Consider

Use the pattern, not a single vessel.

Open the specimen image online

Observe the image at the link

Diffuse-porous wood: makamba vessels

Observe

Compare vessel sizes within an increment.

Locate boundaries

Fibres and parenchyma also provide clues.

Compare

Compare vessel distribution with ring-porous wood.

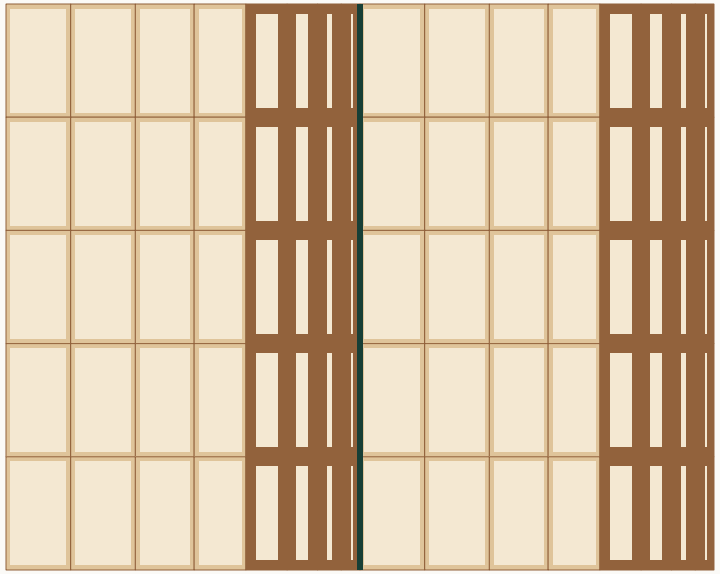
Open the specimen image online

Observe the image at the link

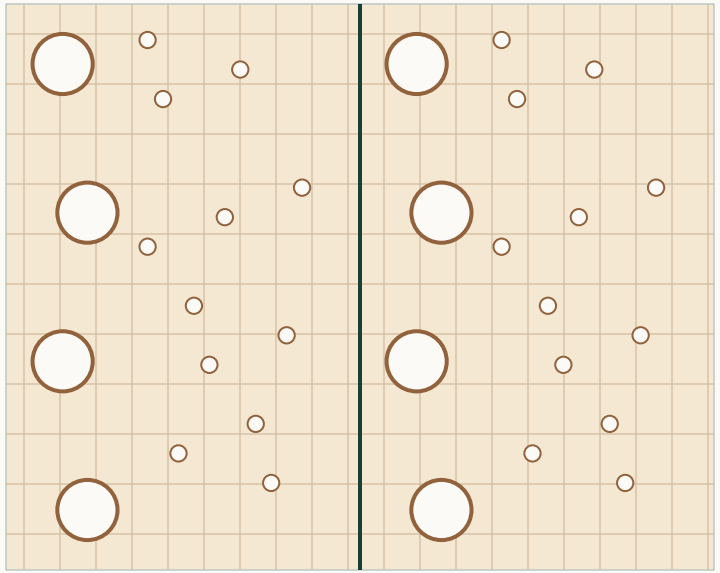
FFPRI: specimen images and record

Compare three wood types

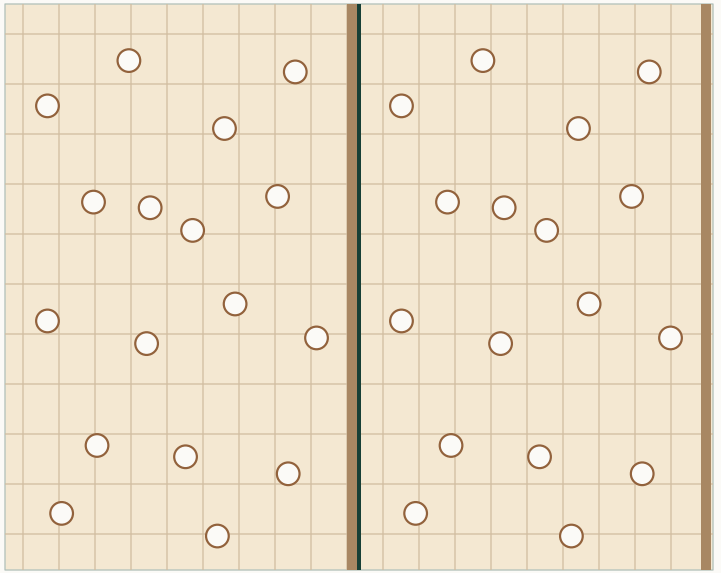
Softwood



Ring-porous



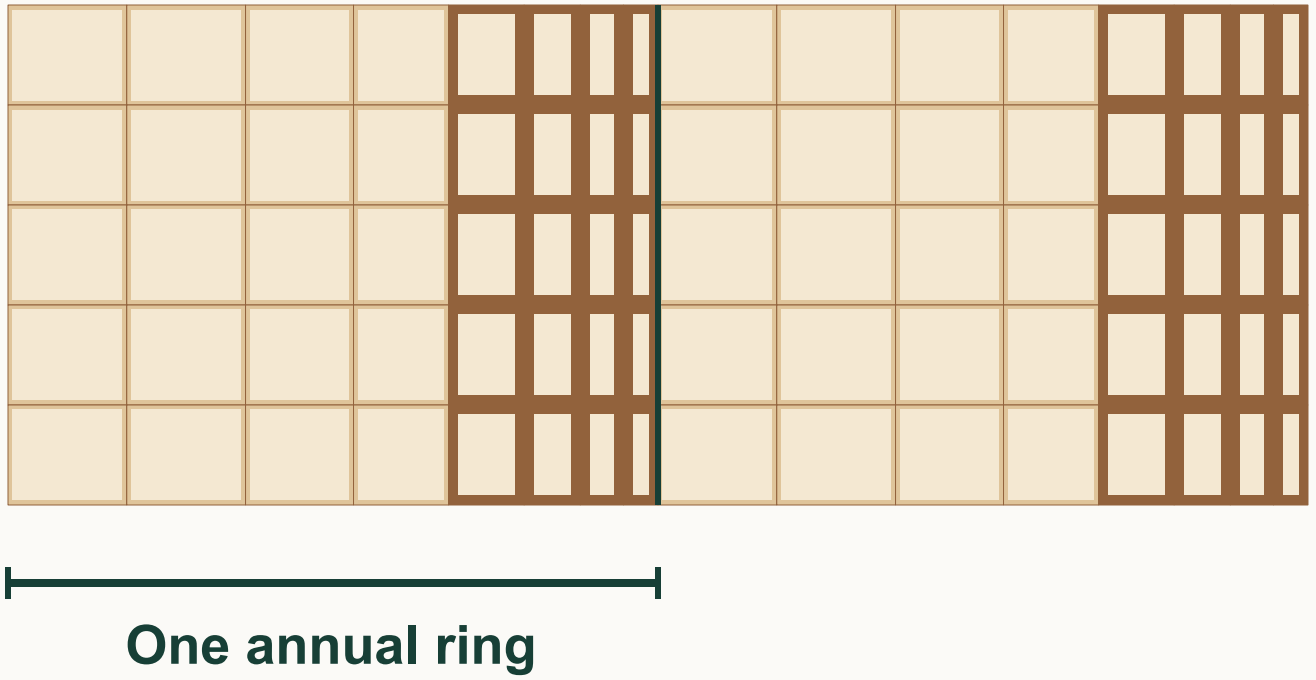
Diffuse-porous



Boundary criteria differ, even when all samples are transverse sections.

Original schematic; not a specimen or measurement, and not to scale.

A ring is an increment of wood



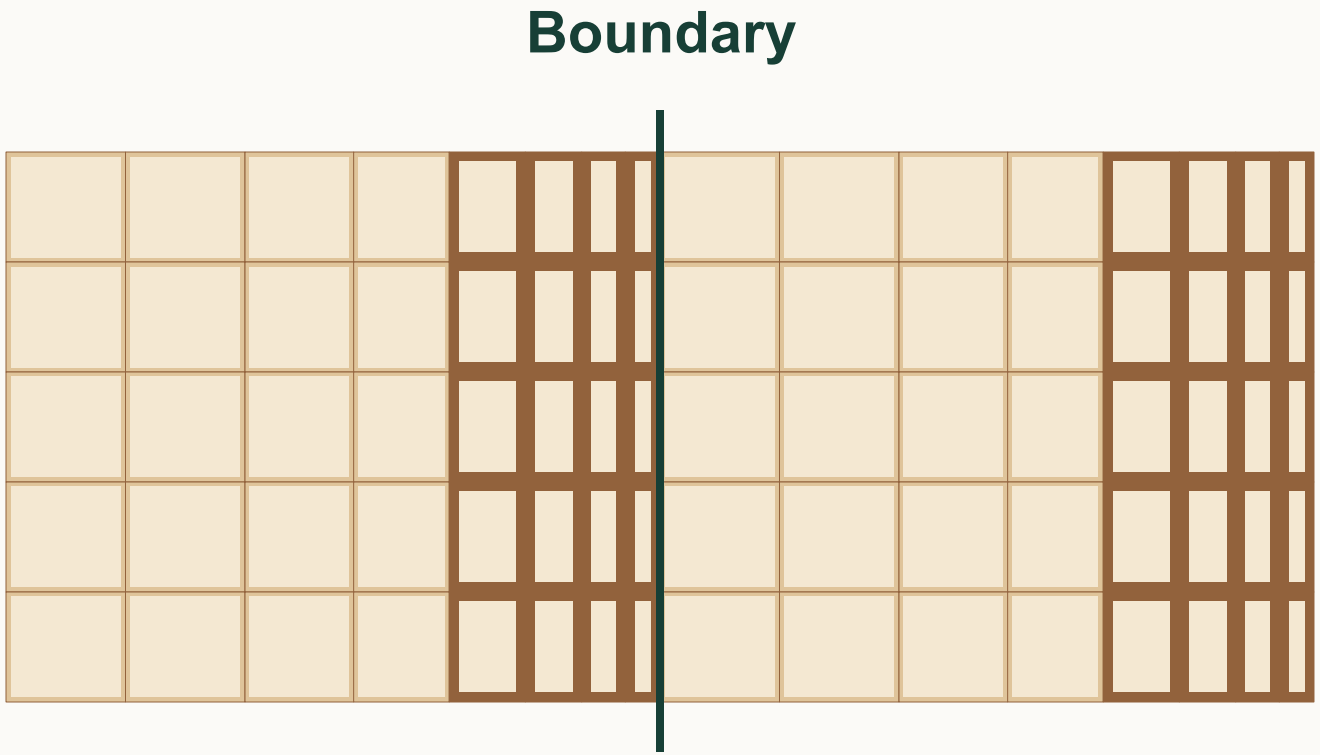
Original schematic; not a specimen or measurement, and not to scale.

A yearly increment includes earlywood and latewood.

A tree-ring boundary separates adjacent annual increments.

A dark band alone is not the whole yearly increment.

Where should the boundary be placed?



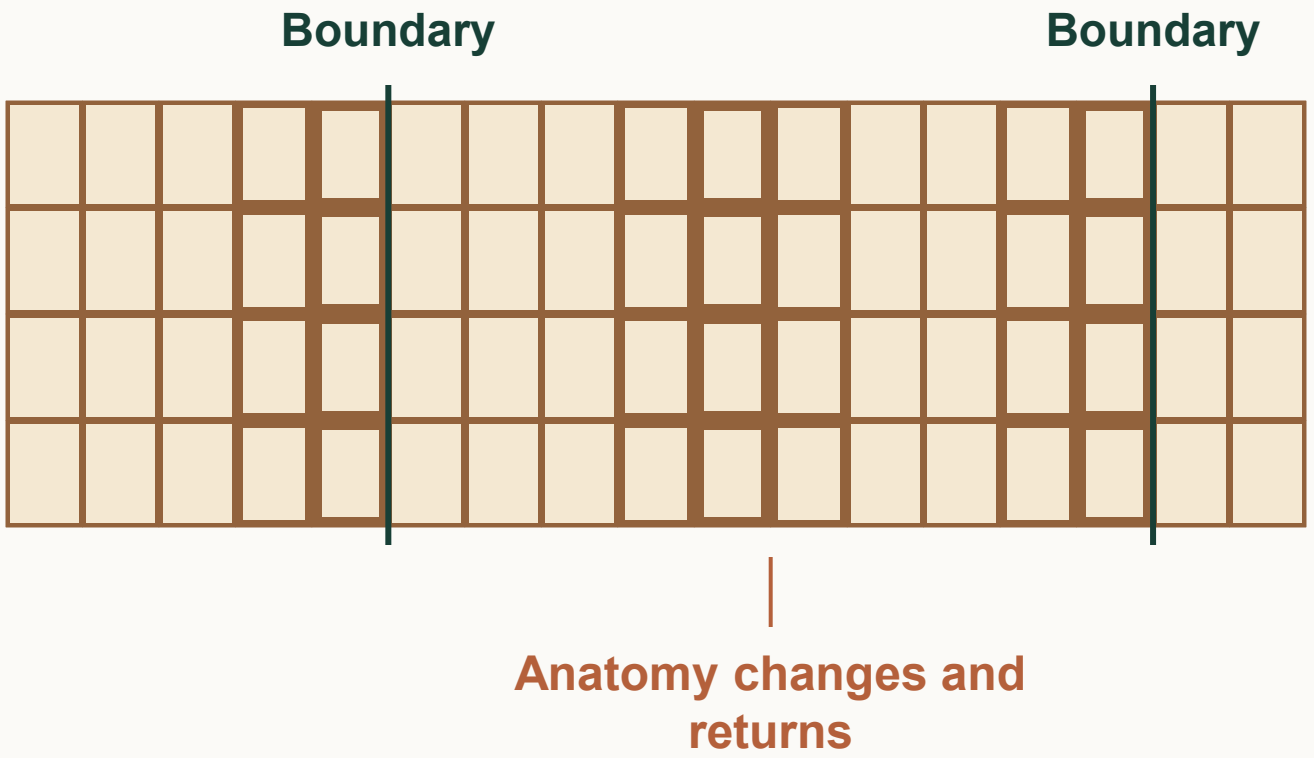
Latewood of the older increment → earlywood of the next

Follow changes in lumen size and wall thickness.

Check anatomy and continuity when colour is ambiguous.

Original schematic; not a specimen or measurement, and not to scale.

False rings: a boundary-like feature within a ring



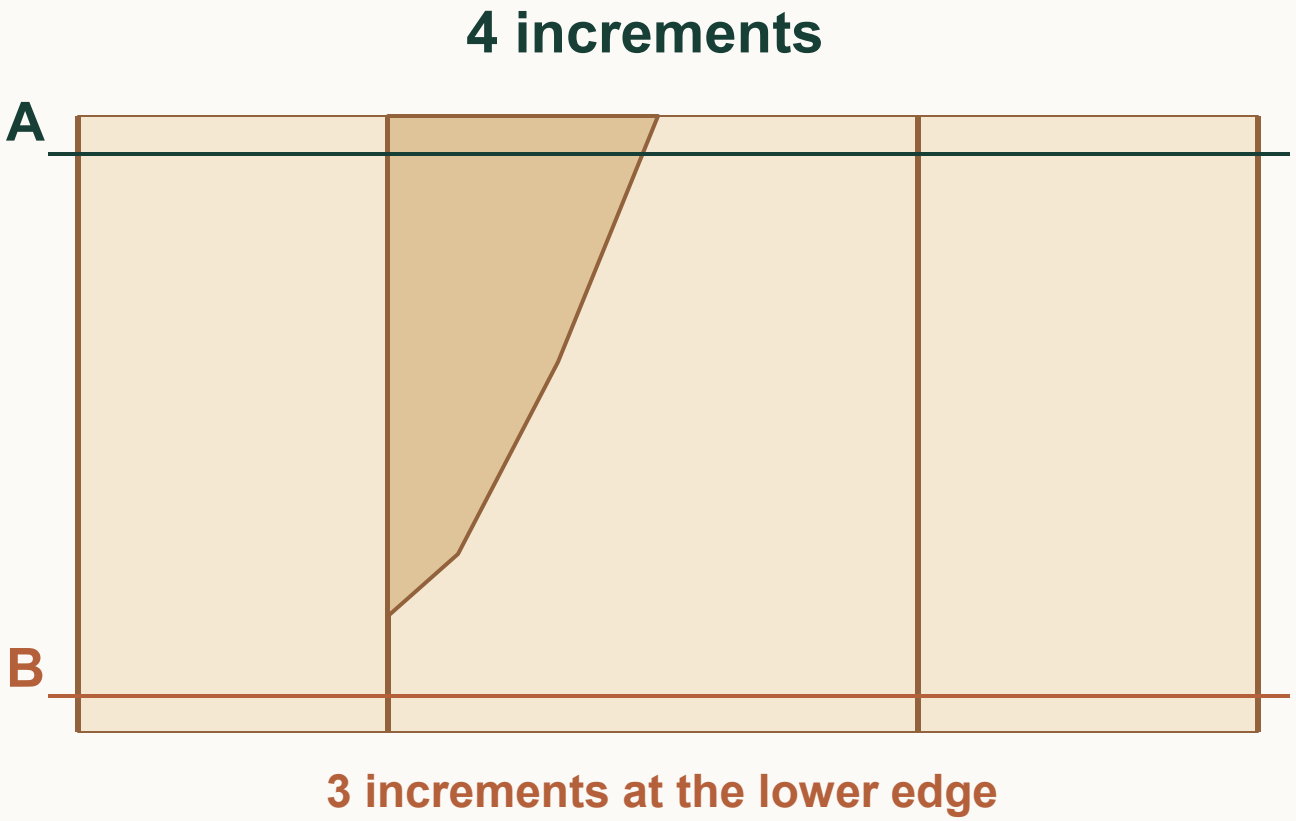
Original schematic; not a specimen or measurement, and not to scale.

False ring

Cell anatomy may change and then return within one growing season.

Check other radii and trees, and verify through crossdating.

Discontinuous rings: an increment pinches out



Discontinuous ring

Visible ring counts can differ between radii of one tree.

A single radius can miss an increment.

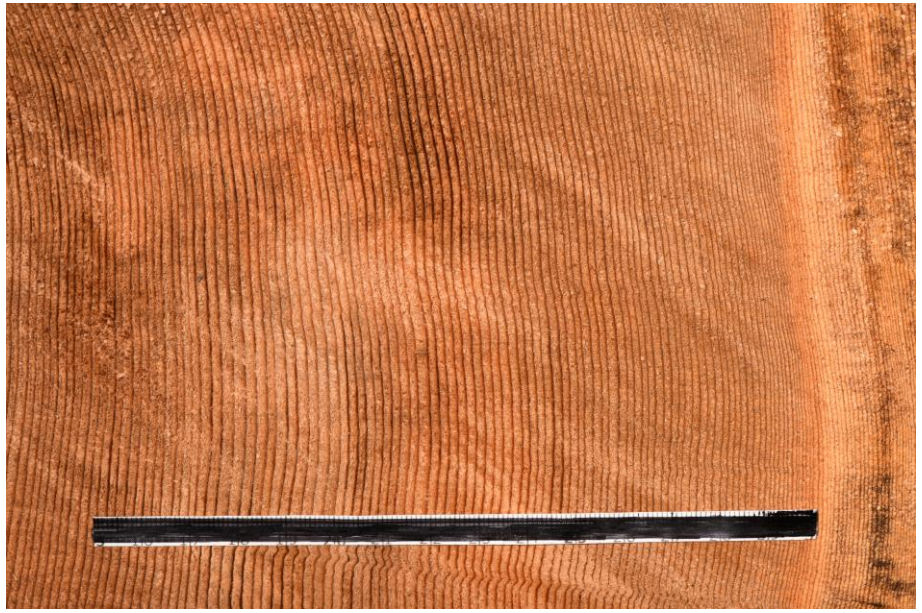
Original schematic; not a specimen or measurement, and not to scale.

Real wood: overview and ring detail

Consider the visible area and scale in each photograph.



A: Cross-section overview



B: Ring detail

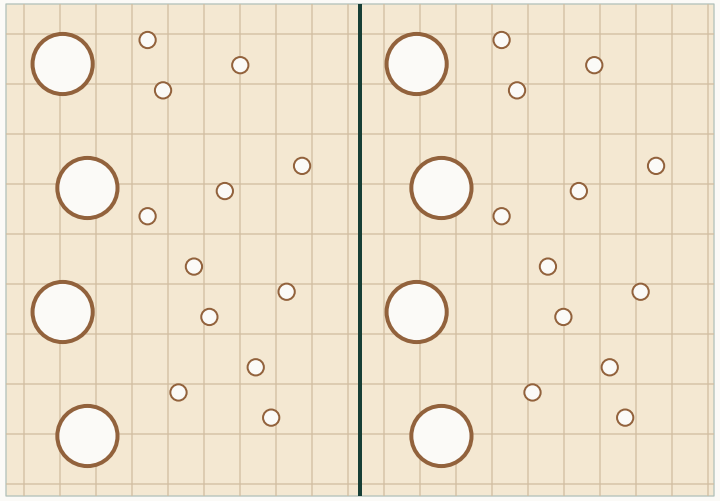
How do ring width and shape vary across the visible wood?

Whether A and B show the same specimen is unconfirmed; photos alone do not date or identify it.

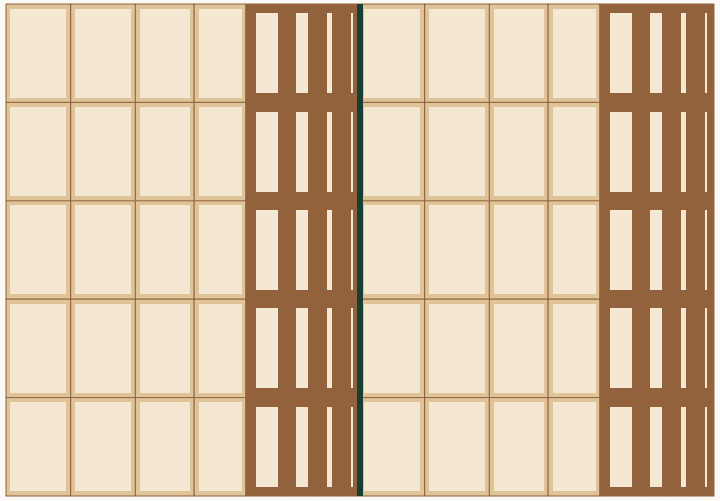
Photos: H. Yonenobu | Yoshino cedar study: Yonenobu et al. (2026)

Observation A: three schematic samples

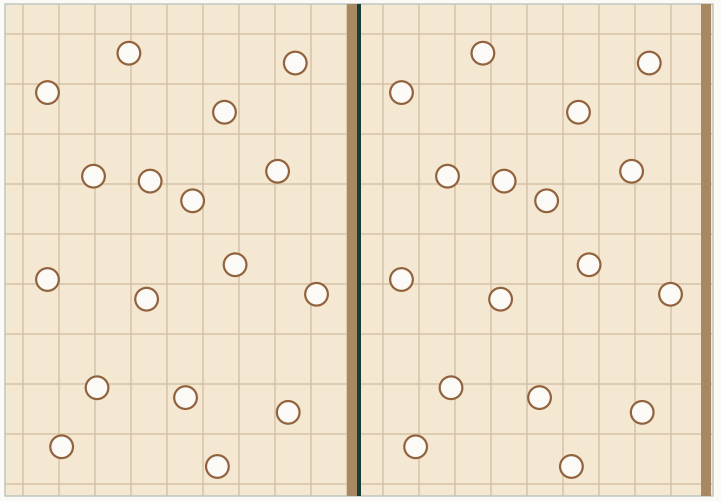
A



B



C

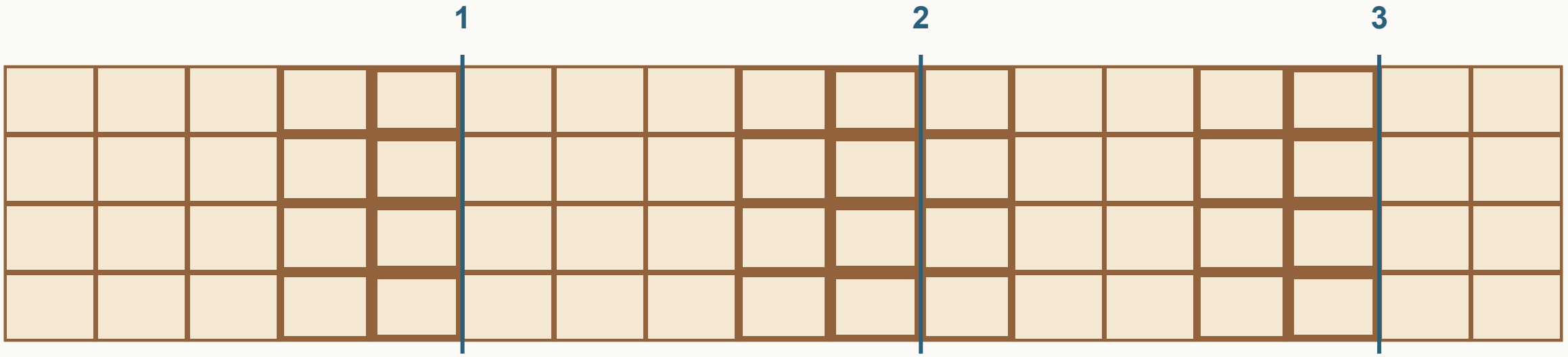


Which wood type is represented by A, B and C?

Mark boundary candidates and give your evidence.

Original schematic observation samples. They cannot identify a species.

Observation B: which boundary separates years?



Compare candidates 1, 2 and 3.

What additional evidence would resolve the interpretation?

Original schematic; not a specimen or measurement, and not to scale.

Discuss observations and interpretations

Sample	Interpretation	Evidence and further checks
A	Ring-porous	Large vessels recur near the start of increments.
B	Softwood	Tracheid lumen size and wall thickness change.
C	Diffuse-porous	Small vessels across increments; examine boundary tissue.
Obs. B	2: false-ring candidate	Anatomy returns; compare other radii and trees.

What can 42 rings tell us?

At 1.3 m: bark present, pith missing.
42 visible rings.

42

This does not establish 42 years since germination.

Assess missing inner wood, years to reach height, annuality and discontinuous rings.

Fictional teaching sample. Height and ring count are not empirical data.

Summary

The cambium adds xylem inward

Wood contains cells involved in transport, support and storage.

Read boundaries from anatomy

Use criteria appropriate to the wood type and plane.

A count alone does not fix a date

Consider sample position, missing wood and ring anomalies.

Real specimens and English resources

Japanese wood anatomy databases	FFPRI
Wood Handbook, Chapter 3	USDA Forest Products Laboratory
Ring-growth anomalies: real examples	University of Arizona LTRR
Glossary of Dendrochronology	WSL
Stems and primary / secondary growth	OpenStax Biology 2e
How do you measure the age of a tree?	NIST

Next: assigning calendar years

Compare growth patterns shared between trees.

Use crossdating to verify the date
of each individual ring.

Preparation: glossary T31–T38 and BG01 statistics.